
Agricultural Development in Paraguay

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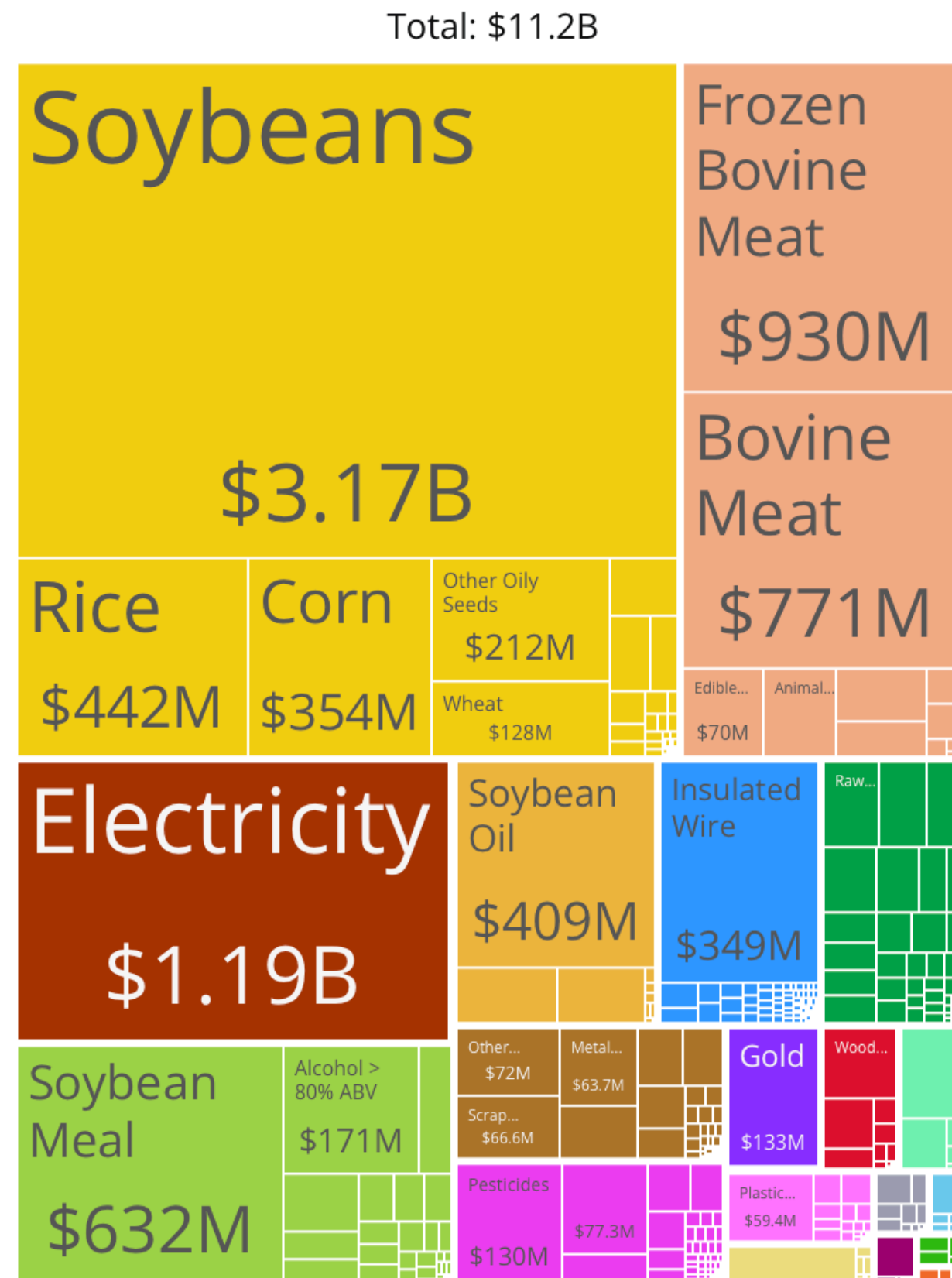


Commodity Boom



THE ASUNCIÓN TIMES · ECONOMY · 4 MONTHS AGO
· 1 MIN READ

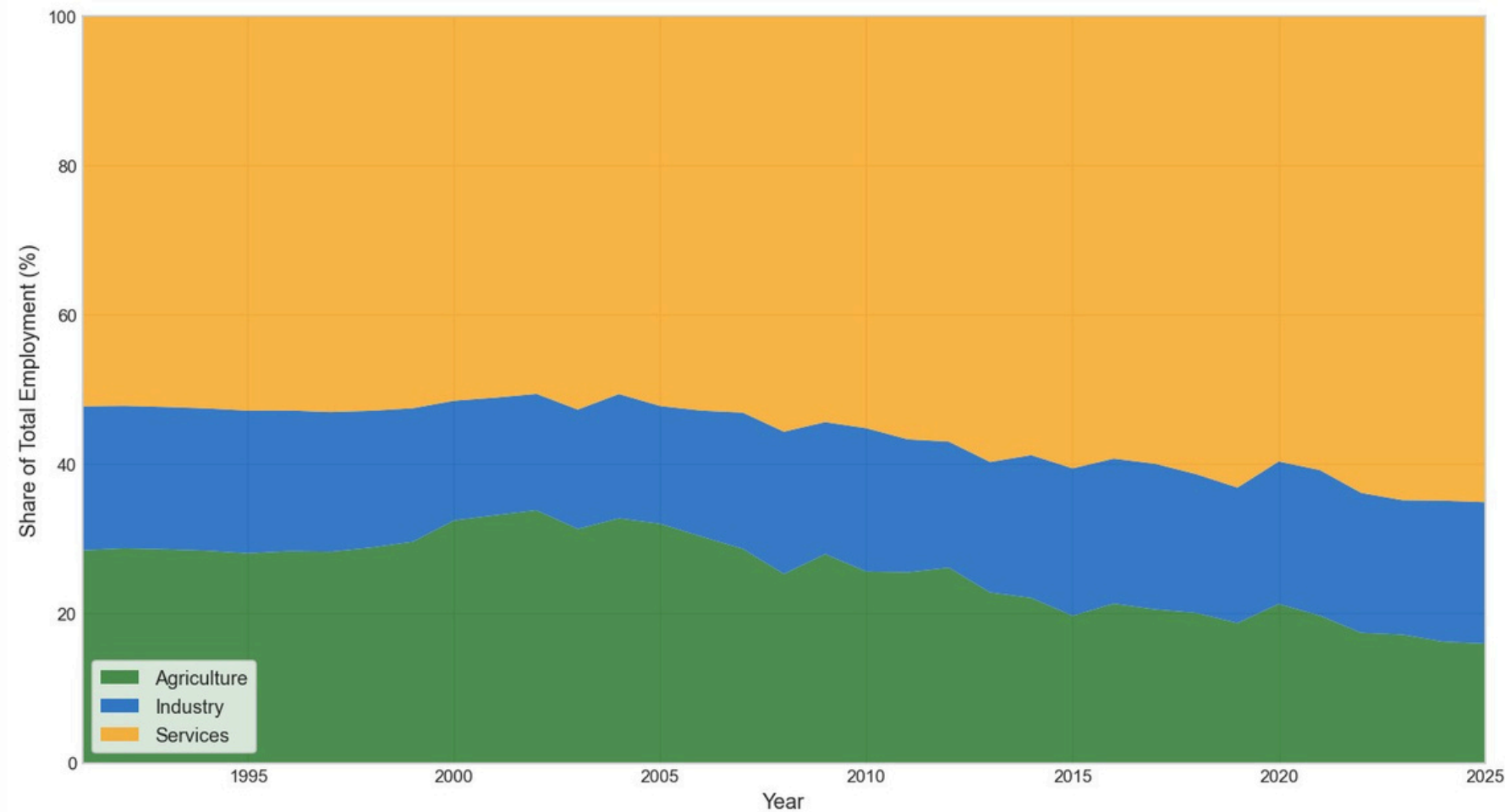
Paraguay's Soybean Industry Closed 2025
With Record Milling And 17% Export
Volume Surge



- Record Surge: Driven by export volume surge and record milling, Paraguay's soybeans production hit a historic peak
- Export Dominance: Agribusiness dominates Paraguay's export value, serving as the primary anchor of the national economy
- Soy & Beef Pillars: The soy complex and beef are the key drivers of Paraguay's foreign exchange earnings

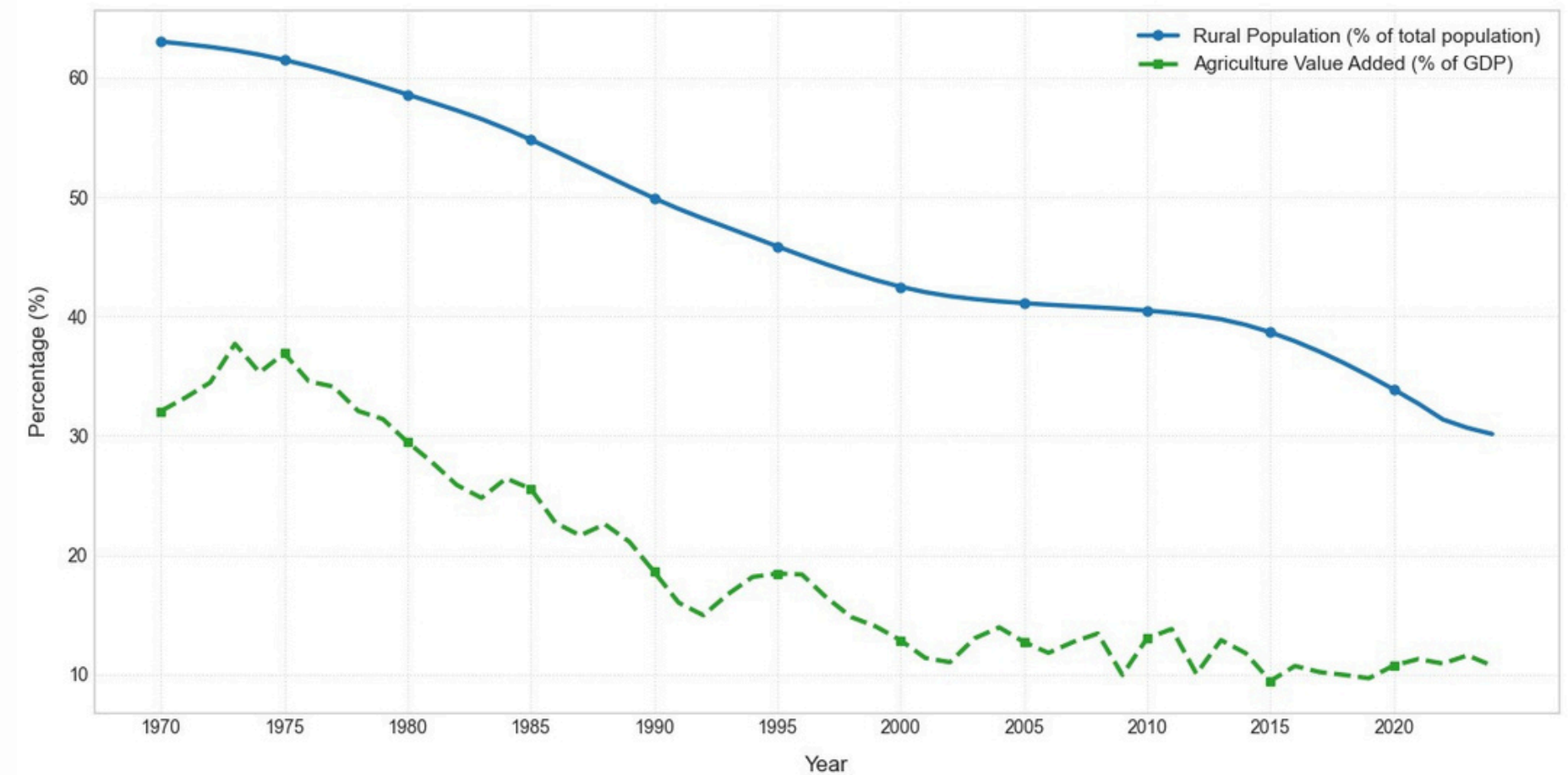
Strutural Transformation

Employment Distribution by Sector in Paraguay (1991-2025)

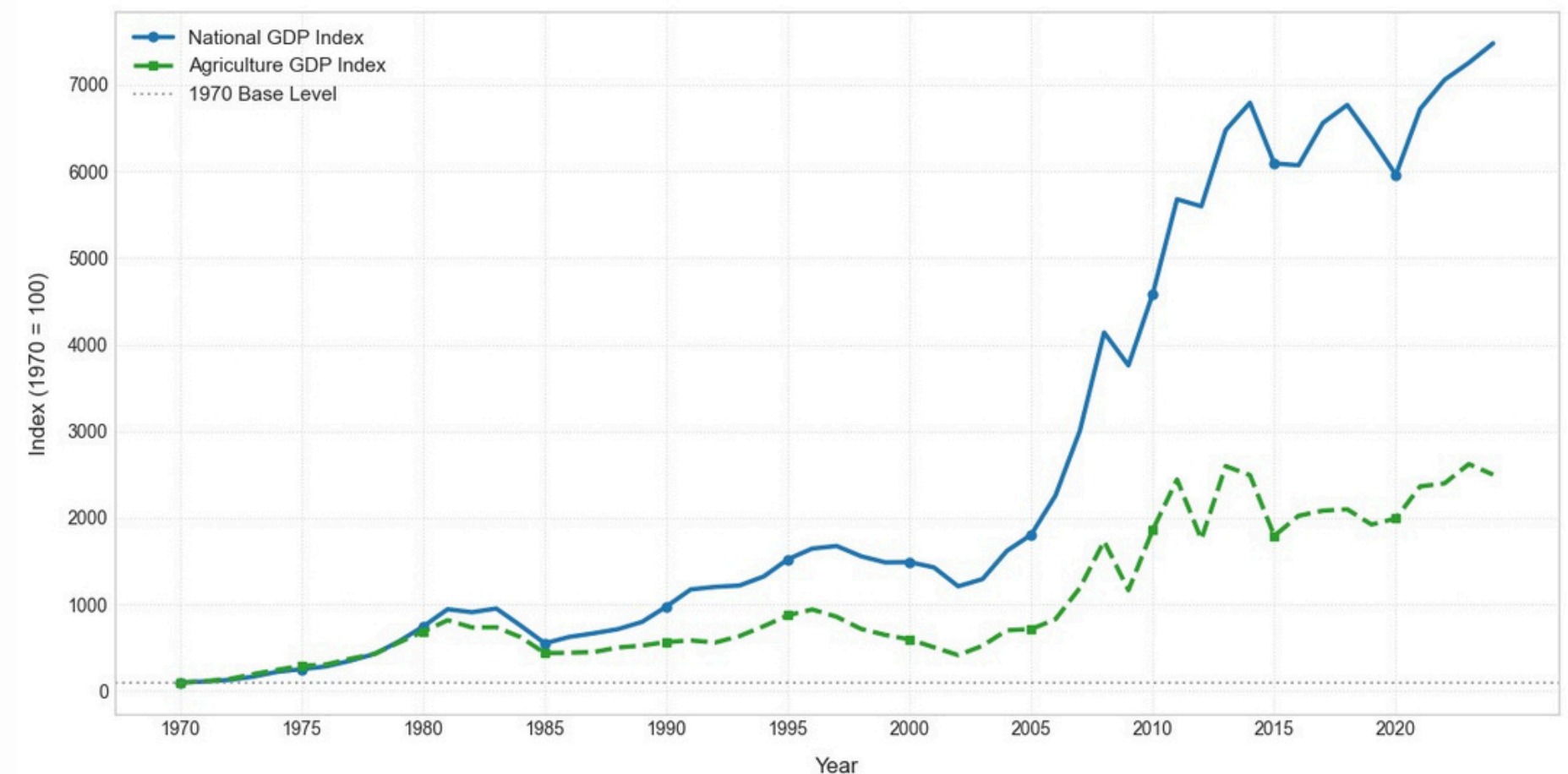


- Demographic & Labor Shifts: Rural population dropped and the agricultural employment share shrank to 16%
- Efficiency Takeoff: While agriculture's share of total GDP decreased significantly, the agriculture GDP index showed nearly 25-fold over the past 55 years, achieving higher economic value with a smaller workforce

Rural Population Share vs. Agriculture Share of GDP in Paraguay (1970-2024)

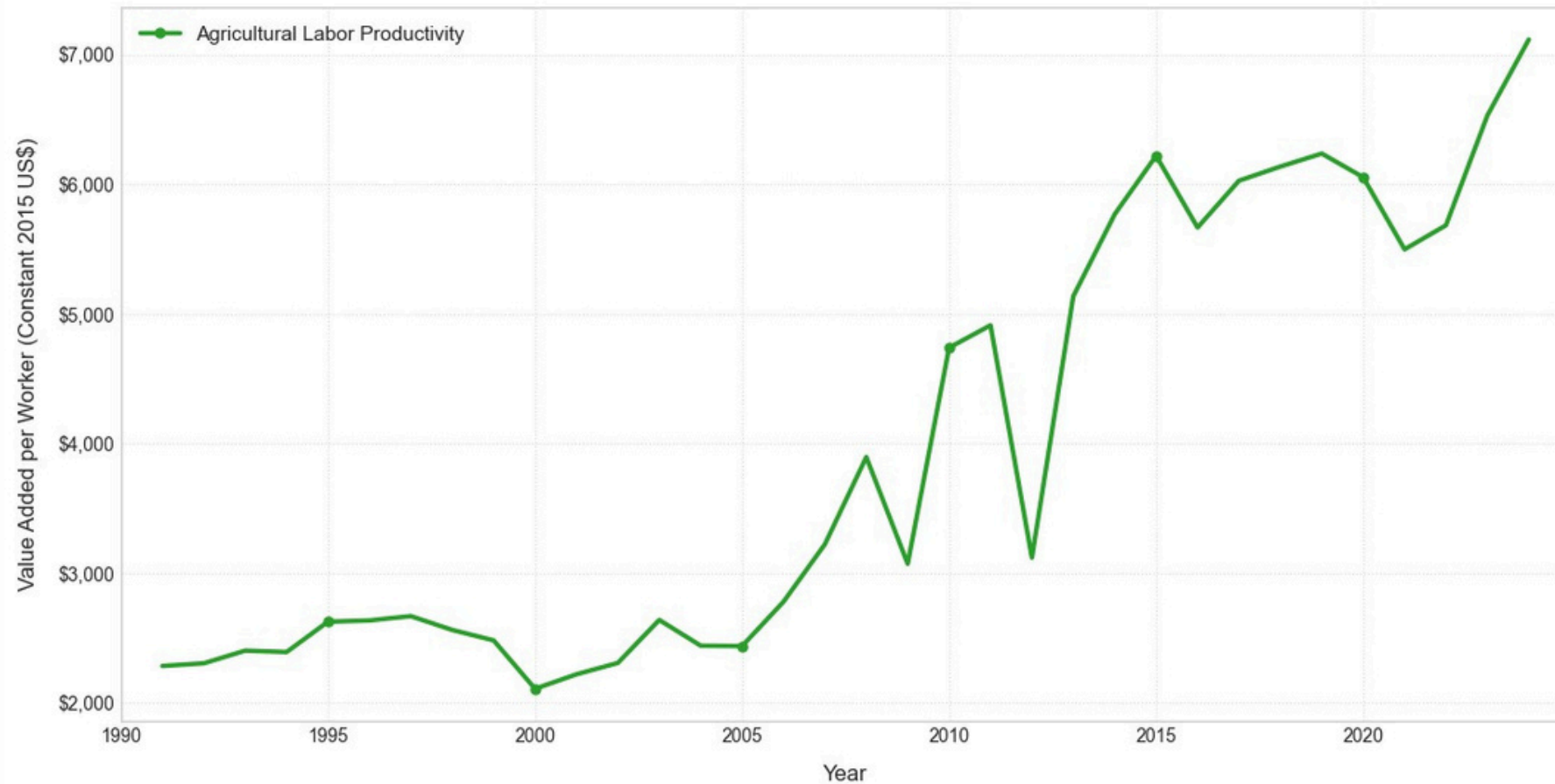


National GDP vs. Agriculture GDP Growth Index in Paraguay (1970-2024)



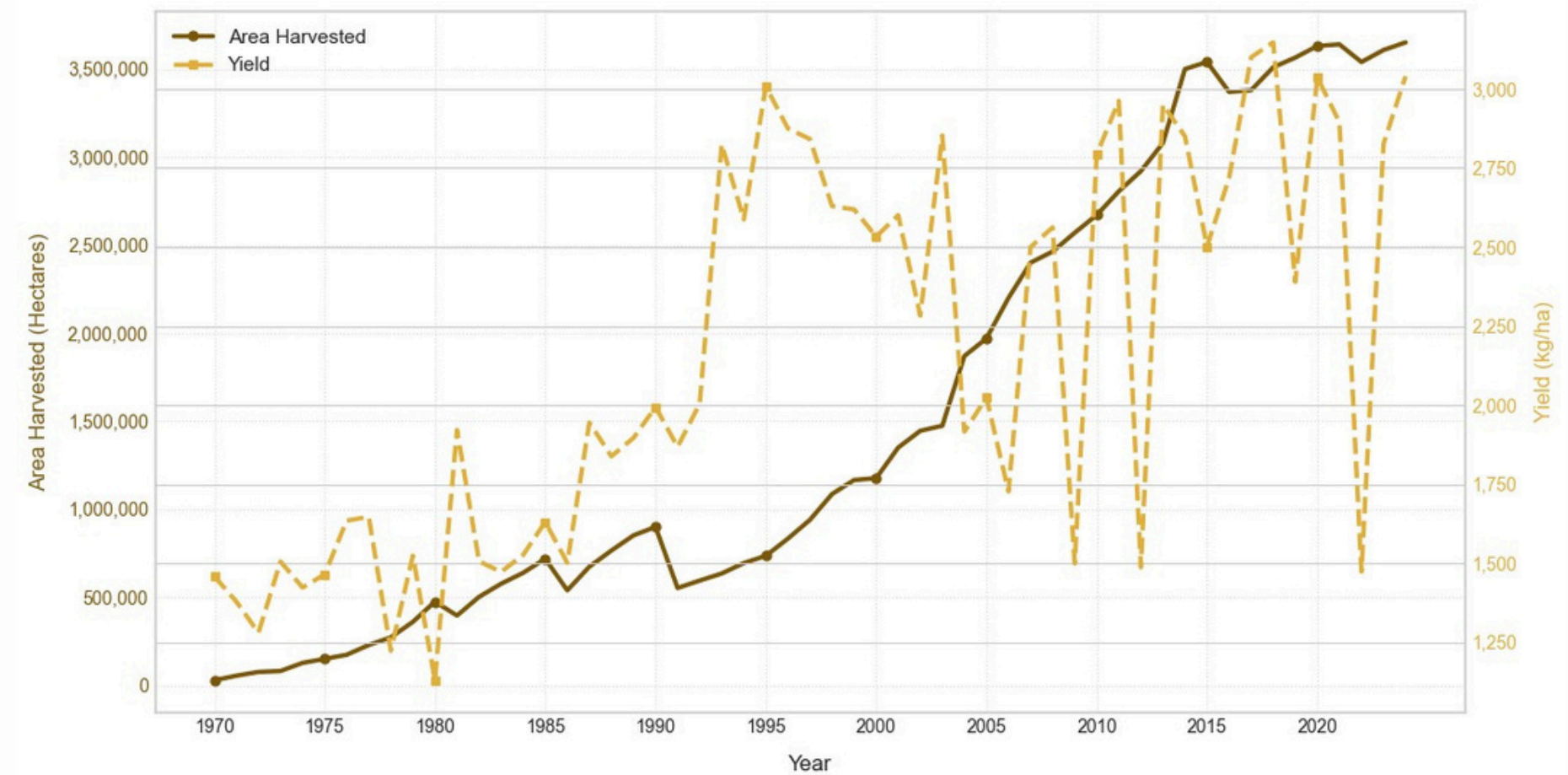
Productivity Surge

Agricultural Labor Productivity in Paraguay (1991-2024)

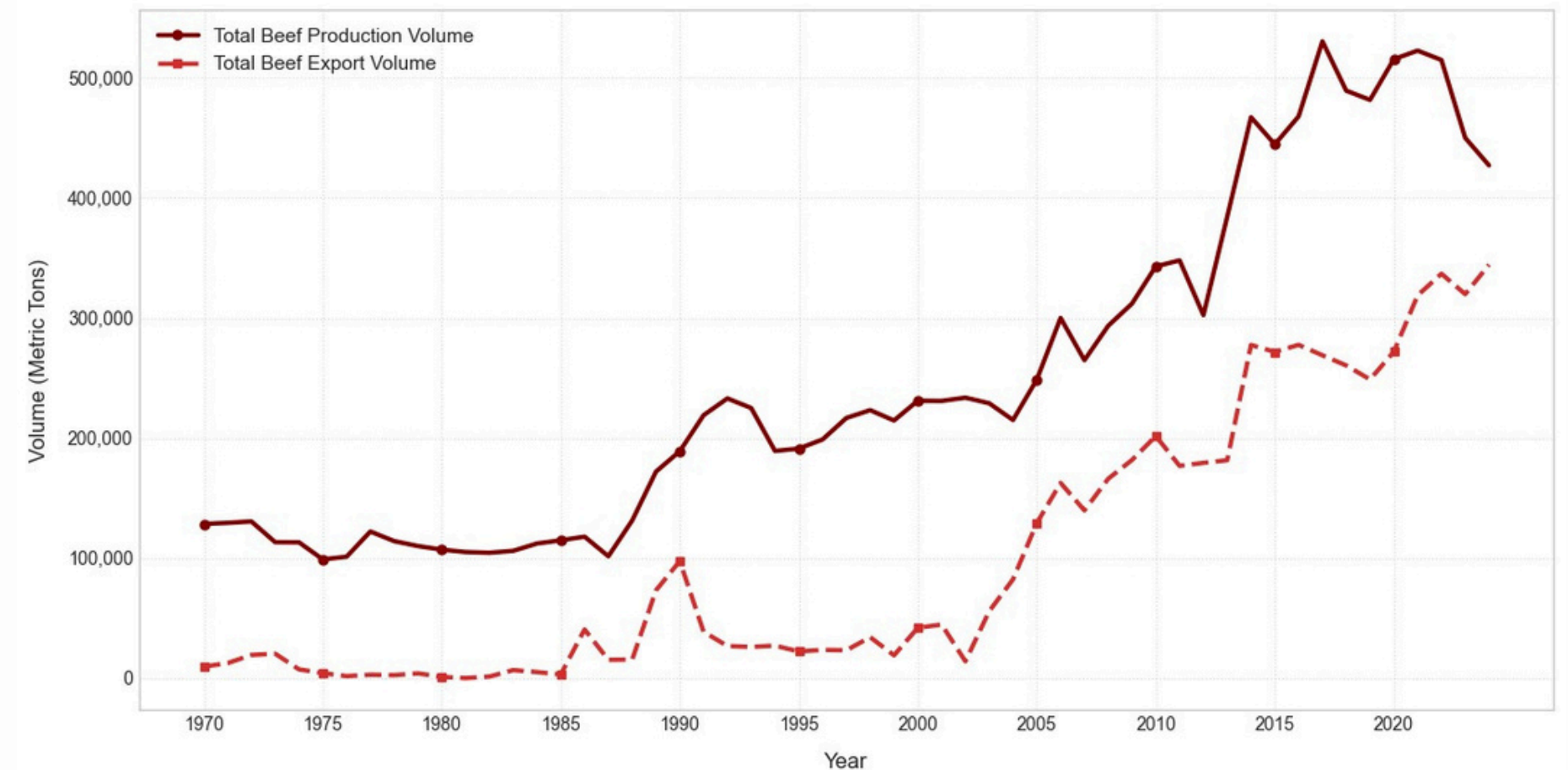


- **Labor Efficiency:** Agricultural labor productivity surged, fueled by mechanization and GM adoption
- **Yield & Area Expansion:** Soybean harvested area surpassed 3.5 million hectares with average yields doubling to around 3,000 kg / ha, growing in pace with booming beef production and exports

Soybeans Area Harvested vs. Yield in Paraguay (1970-2024)

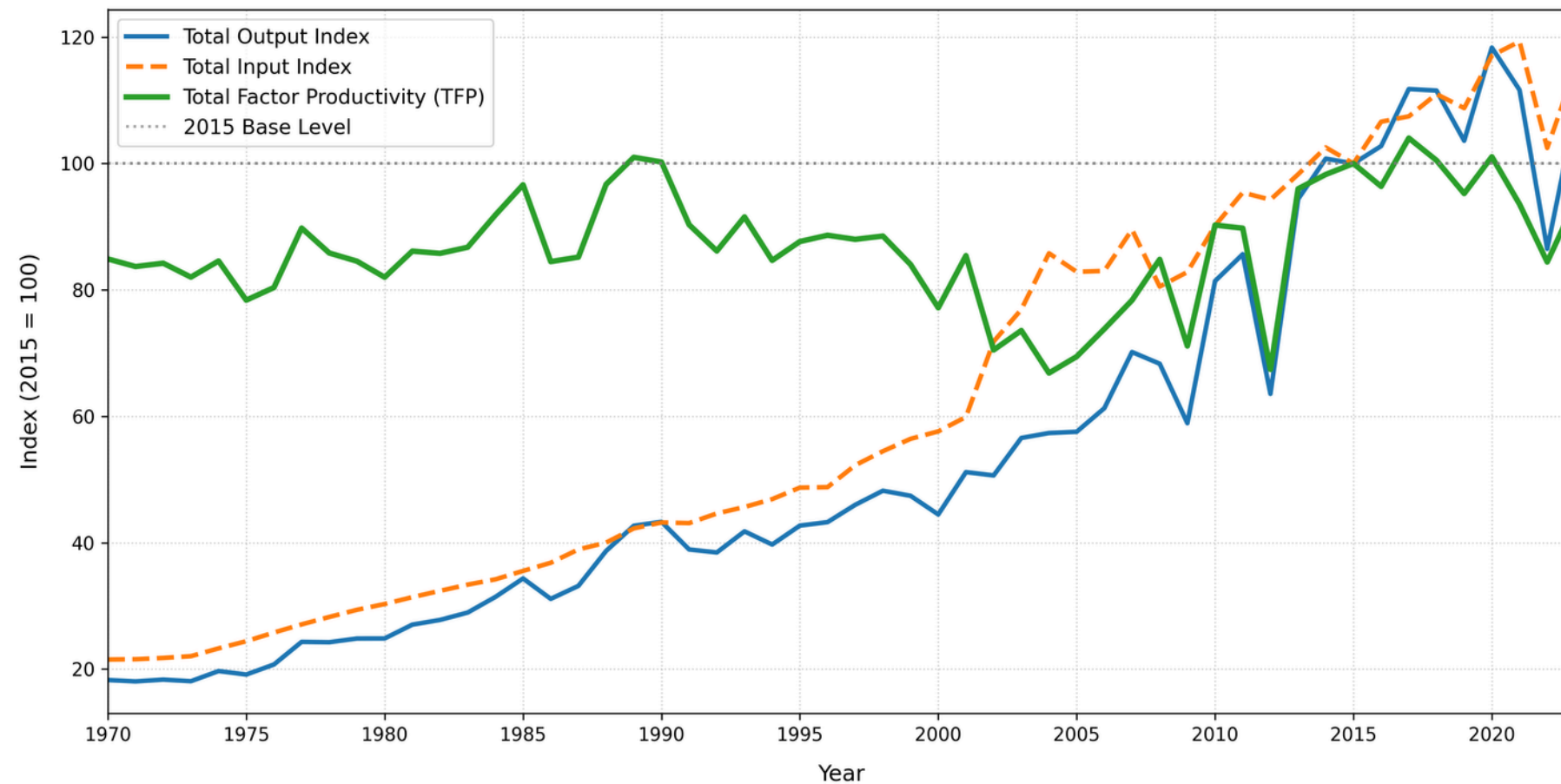


Beef Production Volume vs. Export Volume in Paraguay (1970-2024)

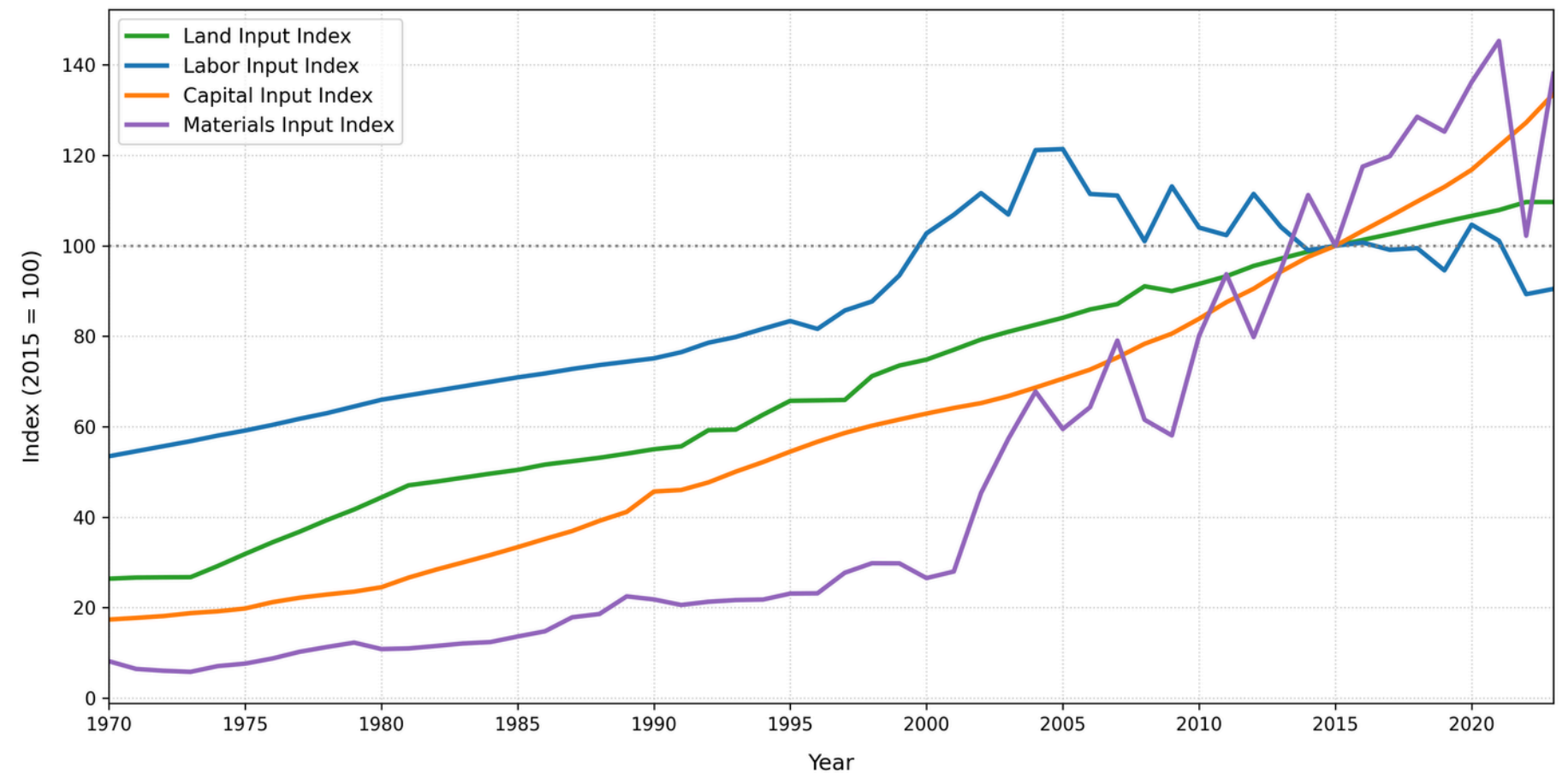


Inputs & TFP Trends

Agricultural Productivity Growth in Paraguay (1970-2023)



Agricultural Input Factors Growth in Paraguay (1970-2023)

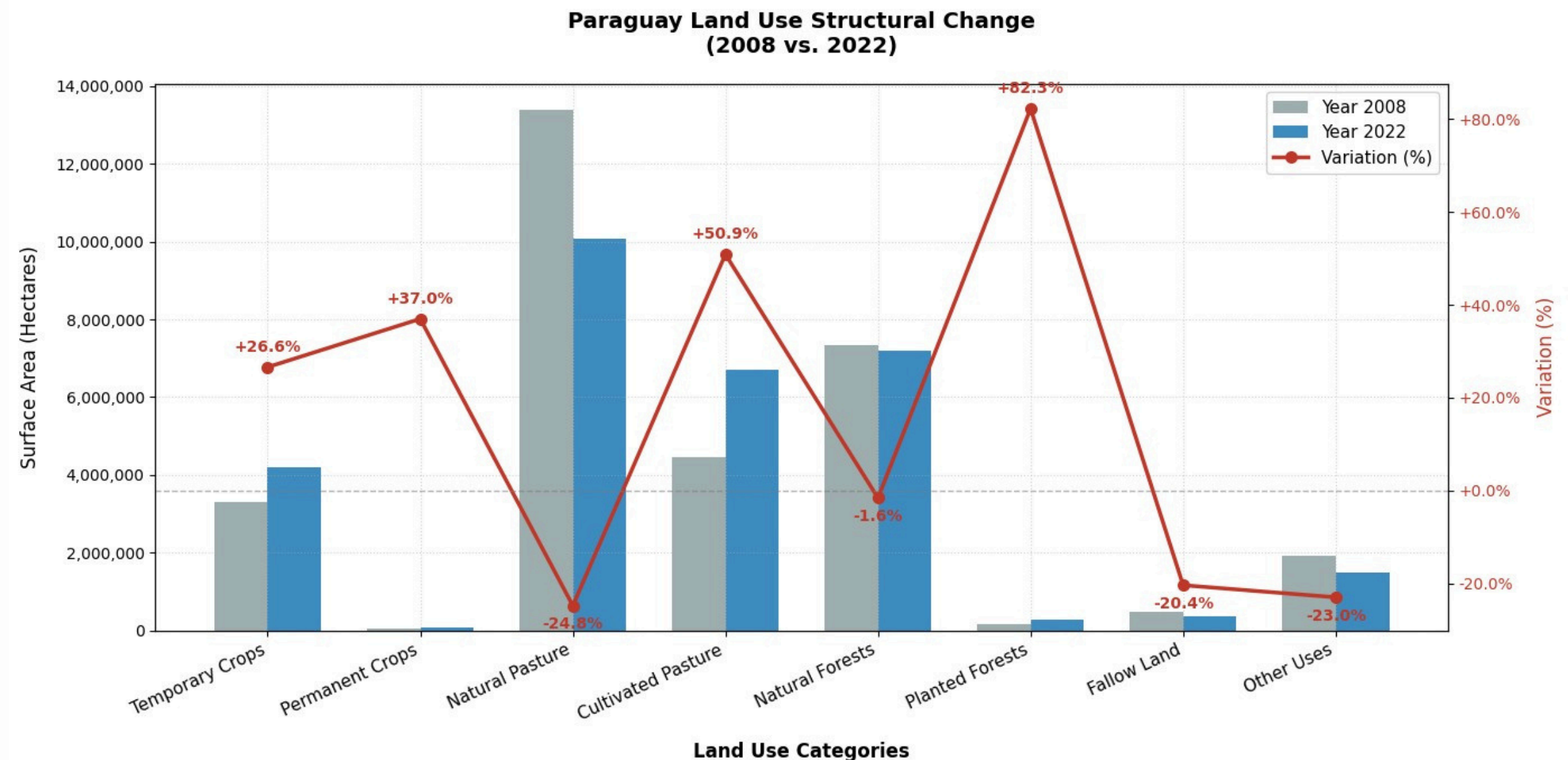


- **TFP Ceiling:** While long-term Total Factor Productivity (TFP) shows a slightly upward trend, its abrupt collapse during weather shocks (e.g. 2022) exposes the limits of this high-input model
- **Resource-Intensive Growth:** Highlight an input-driven model in Paraguay's agricultural development, with a massive growth in capital input and materials input index since 1970
- **Market & Policy Drivers:** The global commodity super-cycle paired with Paraguay's "Triple 10" low-tax regime successfully attracted massive foreign direct investment

Land Concentration & Unequal Distribution

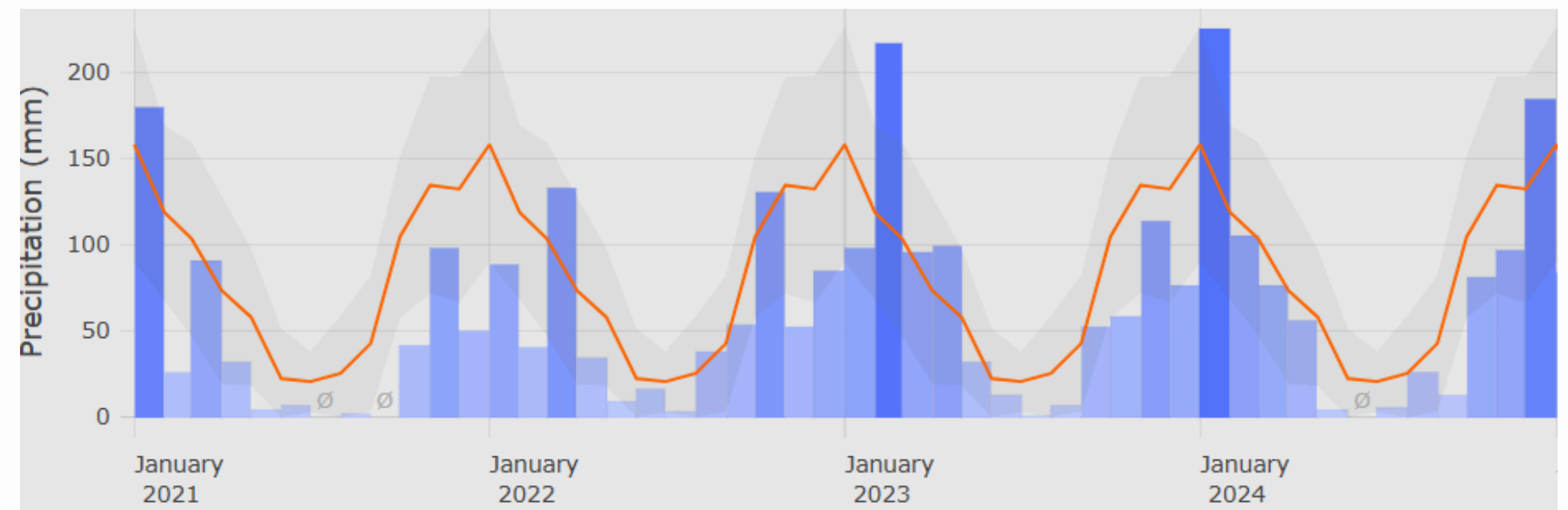
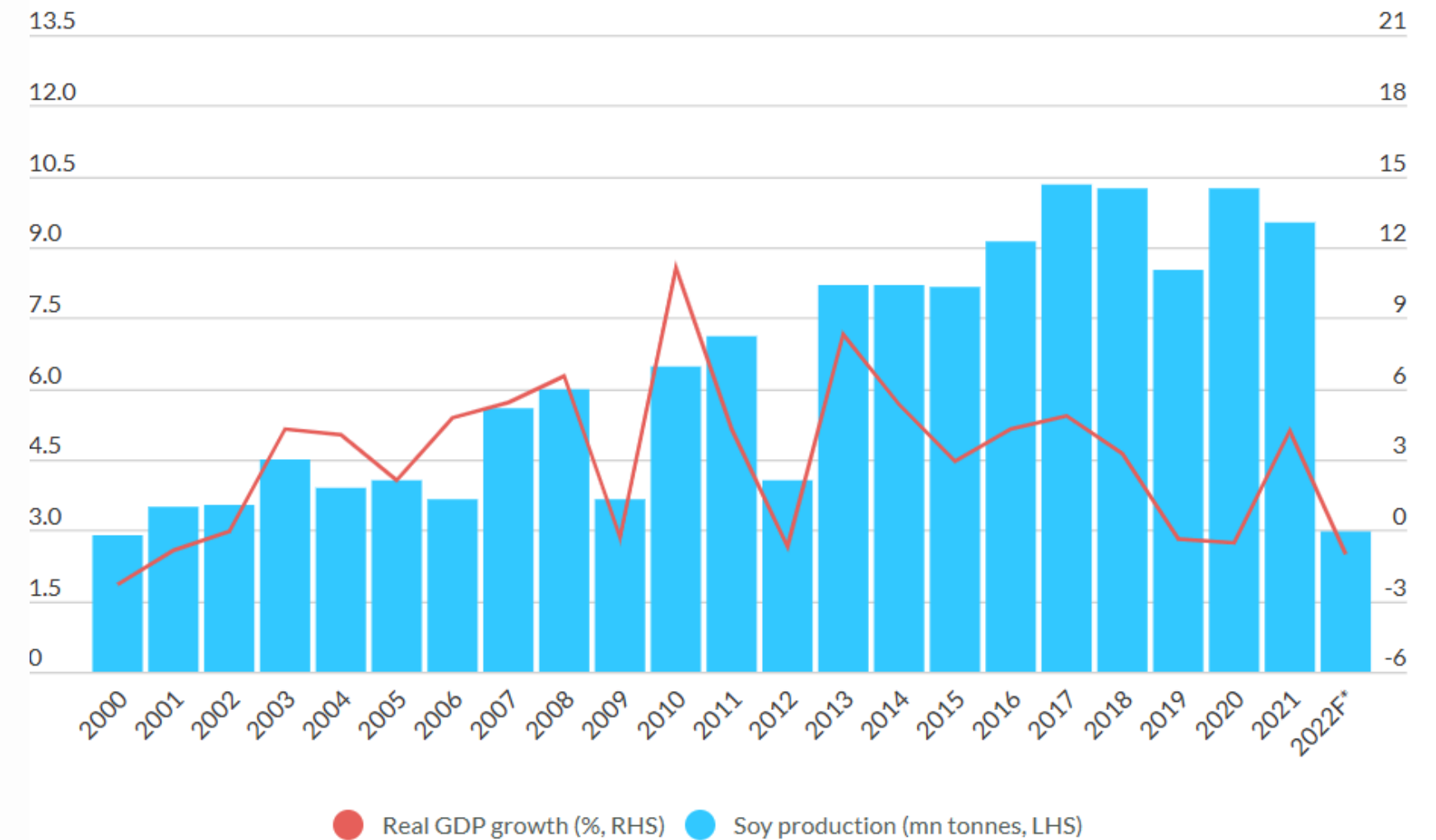
- **Stark Dual Structure:** The 2022 census revealed that few large estates controlled almost 80% of agricultural land, while most smallholders shared just 2.1%
- **Land Consolidation:** Temporary crop area and cultivated pasture expanded by 26.6% and 50.9% from 2008 to 2022
- **Environmental Strain:** Fallow land decreased by 20.4%, indicating continuous, high-intensity land exploitation at the expense of soil health

Farm Size Category	Farms Count	Farms Share (%)	Occupied Area (ha)	Area Share (%)
Smallholder Farms (< 10 ha)	187,172	64.21%	638,737	2.10%
Medium-Sized Farms (10 - 1,000 ha)	99,801	34.24%	6,299,097	20.72%
Large Estates / Latifundia (≥ 1,000 ha)	4,524	1.55%	23,463,826	77.18%
Total Paraguay (2022)	291,497	100.00%	30,401,660	100.00%

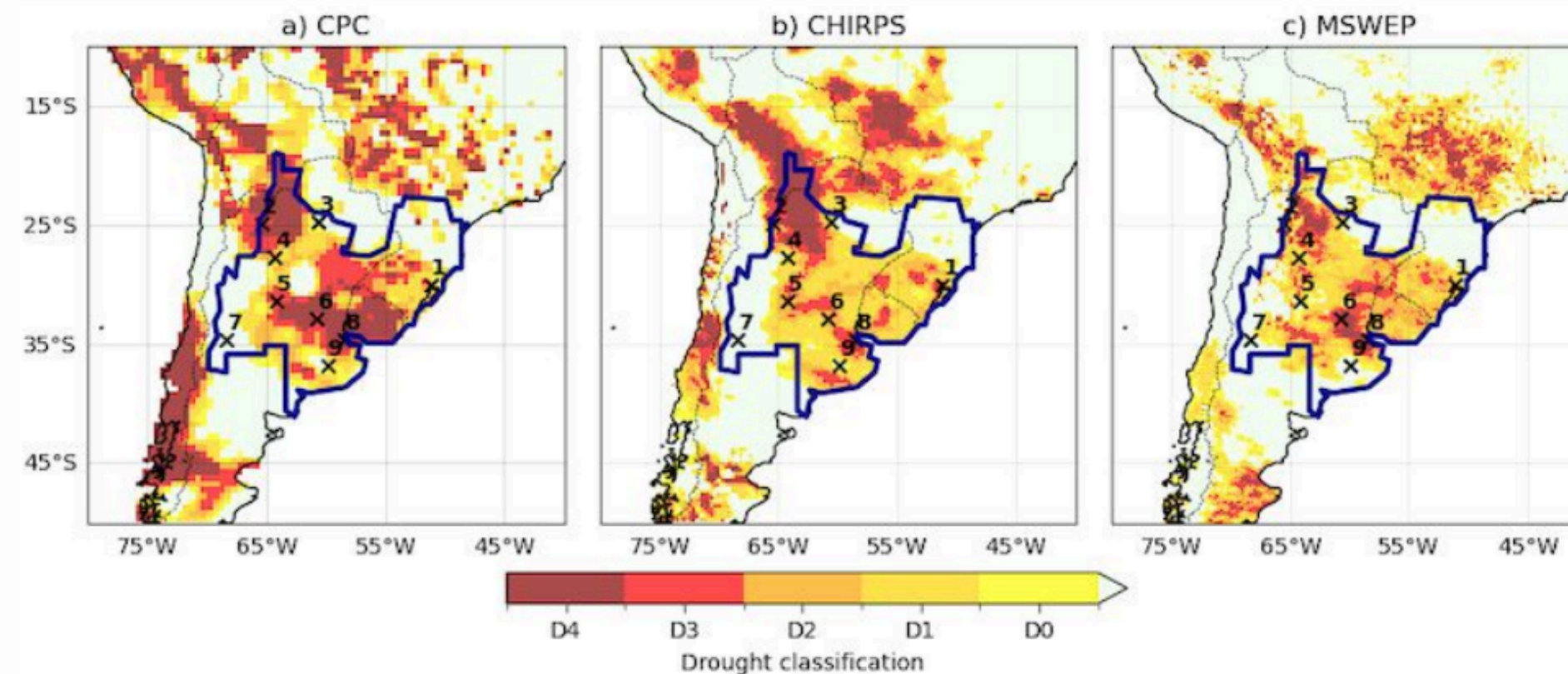
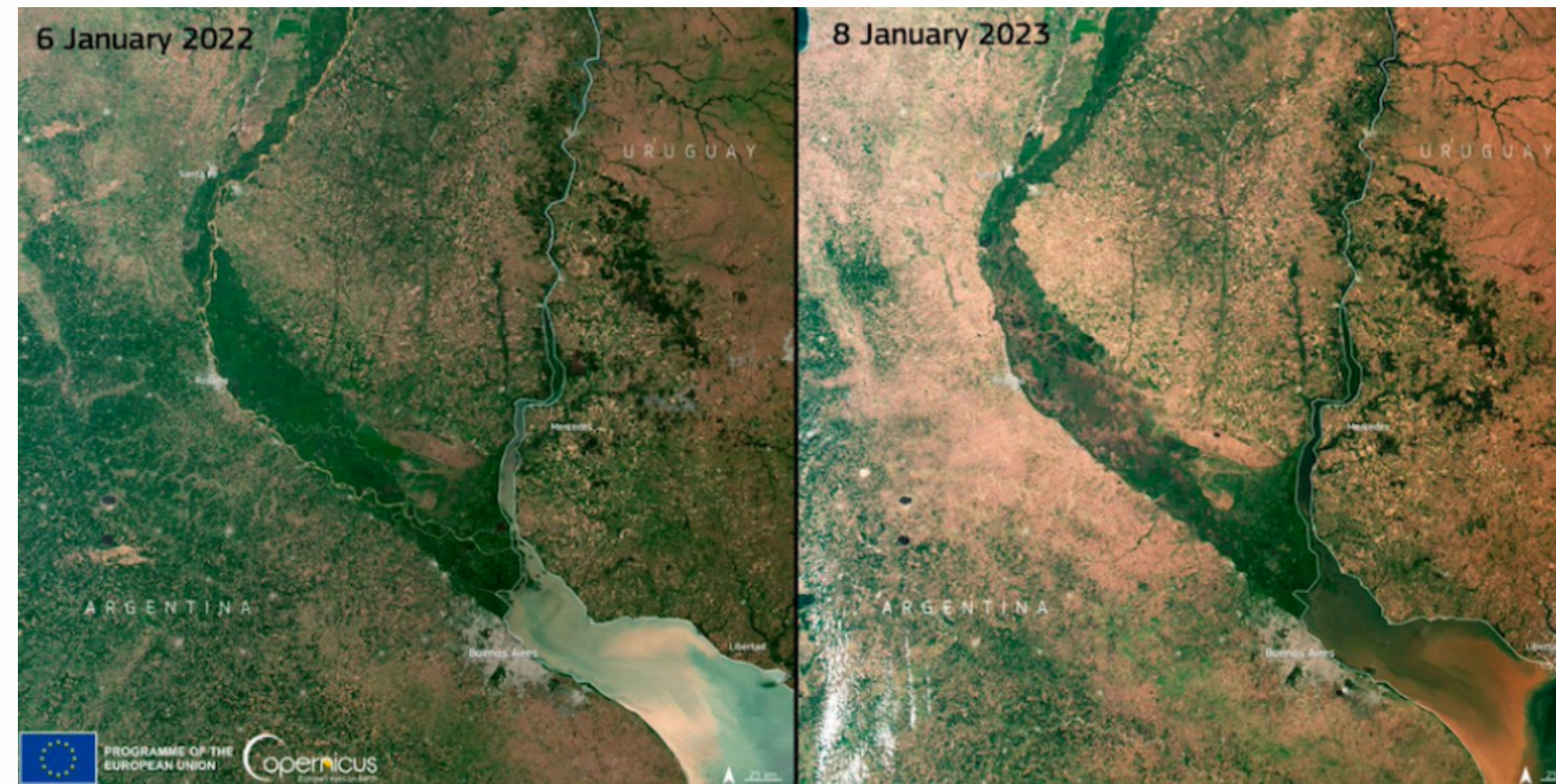


Climate Vulnerability & Macroeconomic Shocks

- Critical Rainfall Timing: As shown in the monthly precipitation data, a severe lack of rain in January 2022 destroyed crops during their flowering stage
- Macroeconomic Sensitivity: The high correlation between soy output and GDP shows that climate shocks are national economic shocks
- Increasing Frequency: Historical trends indicate that extreme drought and forest fires are no longer anomalies but a recurring threat



Climate Vulnerability & Macroeconomic Shocks



Policy Recommendations

Policy 1: Land Tenure Reform & Formalization

Paraguay's extreme land concentration prevents 90% of producers from accessing formal financial systems.

Action:

- Implement a National Digital Land Registry to resolve overlapping titles and secure formal ownership for smallholders.
- Simplify the legal process for "Title Formalization" specifically for farms under 20 hectares.

Expected Impact:

- Financial Inclusion: Secured land titles allow farmers to use land as collateral, unlocking access to formal bank loans.
- Investment: Tenure security encourages long-term investment in soil health and fixed infrastructure.

Policy Recommendations

Policy 2: Climate-Resilient Infrastructure & Crop Diversification

The 2022 drought caused a 60% drop in soybean yield, paralyzing the macroeconomy.

Action:

- Small-scale Irrigation: Government subsidies for drip irrigation and solar-powered water pumps, targeting regions with high climate risk.
- Horizontal Diversification: Provide technical assistance to promote high-value, drought-resistant crops (e.g. sesame, organic sugar cane, or specialty corn).

Expected Impact:

- Economic Stability: Reduce the "GDP-to-Rainfall" correlation, ensuring more stable national growth.
- Risk Mitigation: Diversify crop portfolios protect farmers from price volatility in global soybean markets.

Data Sources & References



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Q & A



Thank you !

